

Quadratic Approximation of Tetration

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$$\begin{aligned}C(b, h) &= 1 + \frac{2 \ln b}{1 + \ln b} h - \frac{1 - \ln b}{1 + \ln b} h^2 \\&= \frac{\ln b - 1}{1 + \ln b} h(1 + h) + h + 1 \\&= \left(\frac{\ln b - 1}{1 + \ln b} h + 1 \right) (1 + h)\end{aligned}\tag{1}$$